

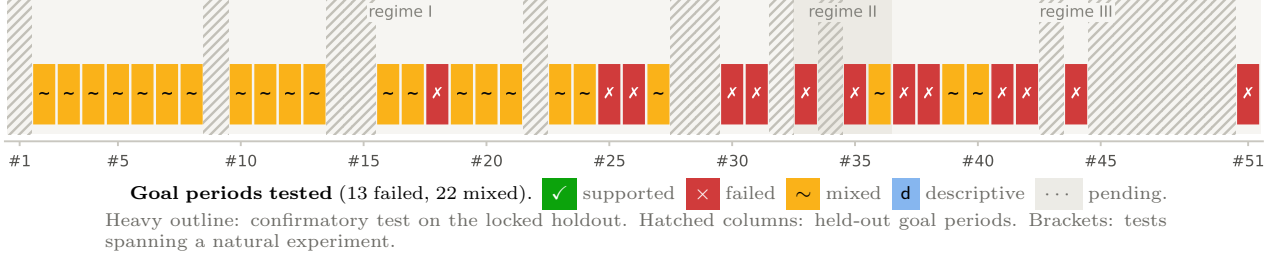
H25: A distance-to-criticality dial, T/T_c , per window and per channel

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1.5/5 Usefulness 2.5/5

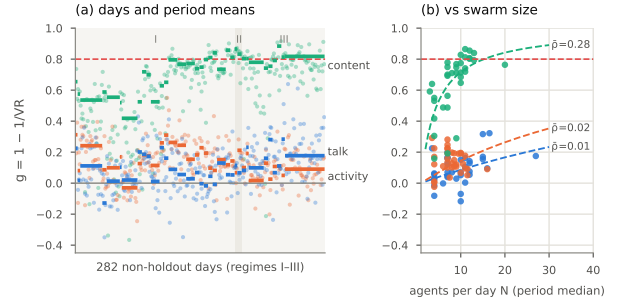
Bottom line. A daily Curie–Weiss loop-gain dial is computable from logs with honest intervals and shows activity and talk far from criticality ($g \approx 0.1$), but it is blind to delayed coupling, mostly noise day to day, and its near-critical content reading is swarm size times a shared-field correlation.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Built a reusable daily dial: the Curie–Weiss loop gain $g = 1 - 1/VR$ and $T/T_c = 1/g$, per day and channel, from logs, with platform stalls masked, slow drives removed and calibrated 90% intervals. Ran it on 282 non-holdout days (35 periods). **Activity and talk are far from criticality on every day:** median $g = 0.10$ and 0.12 ($T/T_c \approx 10, 8$), no upper bound reaching 0.8. The dial reproduces H19 per period (activity $\rho = 0.94$) and sits above the Hawkes-mapped gain in 80% of periods. **The content dial reads near-critical** (median 0.70; 33 days with lower bound > 0.8), so the headline prediction P1 failed. Post hoc, that reading is roster size times a constant per-pair correlation ($\bar{\rho} \approx 0.28$), the signature of a shared field. Day-to-day variation is mostly noise (P5 failed). In simulated swarms the dial misses coupling delayed by minutes. Holdout script frozen, not run.



Observables. (a) Daily dial (dots) and period means (bars) for activity (blue), talk (orange) and content (green) over all non-holdout days; red dashes mark $g = 0.8$. Activity and talk stay near 0.1; content climbs once rosters grow. (b) Period dial against swarm size: dashed curves hold the per-pair correlation fixed. Content and activity follow them (field-like), while talk does not rise with N (attention-diluted coupling, as in H19).

PHYSICAL MODEL

Models 01 (Curie–Weiss), 09 (Hawkes), 11 ($O(n)$ spins). $s_i = \pm 1$: agent i active, or posting, in a minute; for content, s_i is its 30-min mean of whitened unit message embeddings ($n = 32$). $h_i(t)$: private fields (schedule, goal), removed by 30-min block means and agent-day means; stalls masked. $q_i = 1 - m_i^2$; g : loop gain; VR: collective over independent variance; $\bar{\rho}$: mean pairwise correlation; N : agents that day.

$$H = -\frac{J_0}{2N} \left(\sum_i s_i \right)^2 - \sum_i h_i(t) s_i, \quad g = \beta J_0 \bar{q}$$

$$\delta M = \frac{\sum_j q_j \beta \delta h_j}{1 - g}, \quad VR = \frac{\text{Var} \sum_i s_i}{\sum_i \text{Var} s_i} = \frac{1}{1 - g}$$

$$g = 1 - VR^{-1}, \quad T/T_c = 1/g \quad (h = 0)$$

$$VR - 1 = (N - 1) \bar{\rho} \Rightarrow g(N) = \frac{(N - 1) \bar{\rho}}{1 + (N - 1) \bar{\rho}}$$

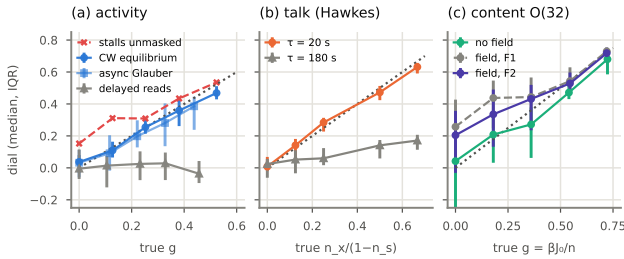
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The useful part is a calibrated instrument and two warnings. Any operator can compute the dial from event logs with `compute_dial`. In this village it says the swarm amplifies a nudge by about 10%: no fast runaway. Warning 1: a green dial does not rule out slow herding. The village's coupling acts through context read minutes later, which the 1-min dial cannot see, as shown on simulated swarms. Warning 2: g rises with headcount at fixed per-pair correlation. “Near-critical” content just means agents in one chat share a topic, and adding agents pushes any field-driven channel toward $g = 1$. Track $\bar{\rho}$, not g , weekly, not daily. As a criticality alarm it is mostly reassurance plus a lower bound.

H25-criticality-dial · generated 2026-10-03 19:21 by in-fra/summaries/build_summaries.py · card hypotheses/H25-criticality-dial/README.md · ratings: H25 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
S1–S4 synthetic recovery (CW, kinetic, stalls, content)	CW and fast Hawkes recovered; stall mask = oracle; delayed coupling invisible; content field removal partial	1 ok, 1 mixed, 2 ×
P1 every day subcritical (all channels)	activity, talk 100%; content 22.5%	×
P2 medians activity 0.03–0.20, talk 0.03–0.25	0.096, 0.122	✓
P3 reproduces H19 ($\rho \geq 0.8$ act., ≥ 0.7 talk); stall drop scales with stall share	0.94, 0.69 (RE means 0.84); drop -0.10 but $\rho = 0.21$	×
P4 H03: talk $\rho \geq 0.3$; LOPO map beats constant; above Hawkes map	0.25 (fast n_x 0.37); SSE 1.78 vs 1.86; 80%	3 of 4
P5 day-to-day signal in $\geq 50\%$ of periods	$\geq 7\%$ (32% unwidened SEs)	×
P6 content $>$ activity; content in 0.15–0.6; F2 lowers it	35/35; 0.70; no change	1 of 3
P7 03–24 switch steps; goal changes do not	$+0.08/-0.06$, CIs span 0; no steps	×/✓



Second figure. Synthetic swarms on 120 real day designs. (a) Equilibrium Curie–Weiss is recovered once stalls are masked; with 2–8 min delayed reads the dial stays at zero. (b) Hawkes talk: fast kernels are recovered, 3-min kernels are not. (c) A time-varying exogenous content field leaks about 0.25; removing operator-message directions (F2) helps little.

GOAL PERIODS IN DETAIL

G02 ~ mixed
G03 ~ mixed
G04 ~ mixed
G05 ~ mixed
G06 ~ mixed
G07 ~ mixed
G08 ~ mixed
G10 ~ mixed
G11 ~ mixed
G12 ~ mixed
G13 ~ mixed
G16 ~ mixed
G17 ~ mixed
G18 × failed
G19 ~ mixed
G20 ~ mixed

+19 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The 1-min dial is a lower bound on the steady-state loop gain: context-mediated coupling is invisible. The content dial is an upper bound: shared time-varying fields survive, and the flat per-pair correlation fits a field. g depends on N , so cross-period contrasts (including the regime-III rise in activity) are partly roster size. Intervals are widened $\times 1.5$ by a synthetic calibration; days with 3–4 agents still get optimistic SEs and pull fixed-effect means. H38’s scheduled-off mask halves the activity dial (0.10→0.05). One embedding model. Not blind to H19 and H03 values; the size-scaling result and the F2tod variant are post hoc.

ROUND 2

Run Stage A of `confirm.py` (content, incl. the size-law test CA4) once committed; build the lagged dial before treating green as safe.

- **Essence.** a trustworthy number for how much the swarm amplifies a push; it must be lag-aware and size-aware, with per-pair correlation as the transferable quantity.
- **H25-R1** A lagged (DC) dial from 1–15 min cross-covariances, validated on the delayed-coupling swarms where the 1-min dial reads 0.
- **H25-R2** Agent-level availability masks from H38’s scaffold states (consolidation, day edges, infra errors).
- **H25-R3** Content fields from real anchors (goal and kick-off embeddings, artifact events) plus an embedding swap; coordinate with H26.
- **H25-R4** Test the size law $g(N) = (N-1)\rho/(1+(N-1)\rho)$ at roster changes and on the holdout (CA4).
- **H25-R5** A weekly dial with change-point detection, at least 5 agents per day.